

APPENDIX A. EXPLICIT RAYLEIGH EQUATIONS

In the nondegenerate case, we group by singularity. For simplicity, we use the shorthands

$$\psi = \psi_{k,\epsilon}^t, \quad b = b(y), \quad \chi = \chi_{*,\delta_0/10}.$$

$$(1) \text{ For } \mathcal{R}_{1,1,\epsilon}^t(y, \lambda) = \left(\partial_y^2 - k^2 - \frac{b''(y)}{b(y) - \lambda + i\epsilon} \right) (\partial_\lambda D_\lambda \psi_{k,\epsilon}^t - \mathcal{S}_{1,1,\epsilon}^t):$$

$$\mathcal{R}_{1,0,\epsilon}^t(y, \lambda) = c_1^{(1,0)}(y) \frac{\log(b(y) - \lambda + i\epsilon)}{b(y) - \lambda + i\epsilon} + \frac{c_2^{(1,0)}(y) + \mathfrak{c}_1^{(1,0)}(y) \partial_y \psi}{b(y) - \lambda + i\epsilon} + c_3^{(1,0)}(y) \log(b(y) - \lambda + i\epsilon) + c_4^{(1,0)}(y), \quad (\text{A.1})$$

ndR10

with

$$\begin{aligned} c_1^{(1,0)}(y, \lambda) &= 0, \\ c_2^{(1,0)}(y, \lambda) &= \left[2b' \partial_y \left(\frac{1-\chi}{(b')^2} \right) + \frac{(1-\chi)b''}{(b')^2} \right] (b'' \psi - \omega_{0k}) + \frac{2(1-\chi)}{b'} (b''' \psi - \omega'_{0k}), \\ \mathfrak{c}_1^{(1,0)}(y) &= \frac{2(1-\chi)b''}{b'}, \\ c_3^{(1,0)}(y, \lambda) &= \partial_y^2 \left(\frac{(1-\chi)b''}{(b')^2} \right) \psi + 2\partial_y \left(\frac{(1-\chi)b''}{(b')^2} \right) \partial_y \psi - \partial_y^2 \left(\frac{1-\chi}{(b')^2} \right) \omega_{0k} - 2\partial_y \left(\frac{1-\chi}{(b')^2} \right) \omega'_{0k} \\ &\quad + \frac{1-\chi}{(b')^2} (k^2 \omega_{0k} - \omega''_{0k}), \\ c_4^{(1,0)}(y, \lambda) &= \chi \frac{b'' \psi - \omega_{0k}}{(b(y) - \lambda + i\epsilon)^2} \end{aligned}$$

$$(2) \text{ For } \mathcal{R}_{2,0,\epsilon}^t(y, \lambda) = \left(\partial_y^2 - k^2 - \frac{b''(y)}{b(y) - \lambda + i\epsilon} \right) (\partial_\lambda^2 \psi_{k,\epsilon}^t(y, \lambda) - \mathcal{S}_{2,0,\epsilon}^t):$$

$$\begin{aligned} \mathcal{R}_{2,0,\epsilon}^t(y, \lambda) &= c_1^{(2,0)}(y) \frac{\log(b(y) - \lambda + i\epsilon)}{b(y) - \lambda + i\epsilon} + \frac{c_2^{(2,0)}(y) + \mathfrak{c}_1^{(2,0)}(y) \partial_y D_\lambda \psi + \mathfrak{c}_2^{(2,0)}(y) \partial_y \psi}{b(y) - \lambda + i\epsilon} \\ &\quad + c_3^{(2,0)}(y) \log(b(y) - \lambda + i\epsilon) + c_4^{(2,0)}(y), \end{aligned} \quad (\text{A.2})$$

ndR20

with

$$\begin{aligned} c_1^{(2,0)}(y, \lambda) &= (1-\chi) \frac{2[b'b''' - 2(b'')^2]}{(b')^4} (b'' \psi - \omega_{0k}), \\ c_2^{(2,0)}(y, \lambda) &= \left\{ -\partial_y^2 \left(\frac{(1-\chi)b''}{(b')^2} \right) - 2b' \partial_y \left[(1-\chi) \left(-\frac{2b'''}{(b')^3} + \frac{3(b'')^2}{(b')^4} \right) \right] - (1-\chi)b'' \left[-\frac{2b'''}{(b')^3} + \frac{3(b'')^2}{(b')^4} \right] \right\} \psi \\ &\quad + \left[4b' \partial_y \left(\frac{(1-\chi)b''}{(b')^3} \right) + \frac{2(1-\chi)(b'')^2}{(b')^3} \right] D_\lambda \psi + (\partial_y^2 - k^2) \left(\frac{(1-\chi)\omega_{0k}}{(b')^2} \right) \\ &\quad - 2b' \partial_y \left[(1-\chi) \left(\frac{2\omega'_{0k}}{(b')^3} - \frac{3b''\omega_{0k}}{(b')^4} \right) \right] - (1-\chi)b'' \left[\frac{2\omega'_{0k}}{(b')^3} - \frac{3b''\omega_{0k}}{(b')^4} \right], \\ \mathfrak{c}_1^{(2,0)}(y) &= \frac{4(1-\chi)b''}{(b')^2}, \\ \mathfrak{c}_2^{(2,0)}(y) &= -2\partial_y \left(\frac{(1-\chi)b''}{(b')^2} \right) + (1-\chi) \left[\frac{4b'''}{(b')^2} - \frac{6(b'')^2}{(b')^3} \right], \\ c_3^{(2,0)}(y, \lambda) &= -(\partial_y^2 - k^2) \left[(1-\chi) \left(\frac{2\omega'_{0k}}{(b')^3} - \frac{3b''\omega_{0k}}{(b')^4} \right) \right] \\ &\quad - \left\{ \partial_y^2 \left[(1-\chi) \left(-\frac{2b'''}{(b')^3} + \frac{3(b'')^2}{(b')^4} \right) \right] + \frac{4k^2(1-\chi)(b'')^2}{(b')^4} \right\} \psi \\ &\quad - \left\{ 2\partial_y \left[(1-\chi) \left(-\frac{2b'''}{(b')^3} + \frac{3(b'')^2}{(b')^4} \right) \right] + \frac{2(1-\chi)b''}{(b')^3} \left[\frac{b'''}{b'} - \frac{2(b'')^2}{(b')^2} \right] \right\} \partial_y \psi \end{aligned}$$

$$\begin{aligned}
& + \left\{ \partial_y^2 \left(\frac{2(1-\chi)b''}{(b')^3} \right) + \frac{2(1-\chi)b''}{(b')^3} \left[\frac{b'''}{b'} - \frac{2(b'')^2}{(b')^2} \right] \right\} D_\lambda \psi \\
& + \left[4\partial_y \left(\frac{(1-\chi)b''}{(b')^3} \right) + \frac{4(1-\chi)(b'')^2}{(b')^4} \right] \partial_y D_\lambda \psi, \\
c_4^{(2,0)}(y, \lambda) = & \chi \left[\frac{2(b''\psi - \omega_{0k})}{(b(y) - \lambda + i\epsilon)^3} + \frac{2b''\partial_\lambda \psi}{(b(y) - \lambda + i\epsilon)^2} \right] - \chi' \frac{2(b''\psi - \omega_{0k})}{b'(b(y) - \lambda + i\epsilon)^2}.
\end{aligned}$$

$$(3) \text{ For } \mathcal{R}_{1,1,\epsilon}^l(y, \lambda) = \left(\partial_y^2 - k^2 - \frac{b''(y)}{b(y) - \lambda + i\epsilon} \right) (\partial_\lambda D_\lambda \psi_{k,\epsilon}^l - \mathcal{S}_{1,1,\epsilon}^l):$$

$$\begin{aligned}
\mathcal{R}_{1,1,\epsilon}^l = & c_1^{(1,1)}(y) \frac{\log(b(y) - \lambda + i\epsilon)}{b(y) - \lambda + i\epsilon} + \frac{c_2^{(1,1)}(y) + c_1^{(1,1)}(y) \partial_y D_\lambda \psi + c_2^{(1,1)}(y) \partial_y \psi}{b(y) - \lambda + i\epsilon} \\
& + c_3^{(1,1)}(y) \log(b(y) - \lambda + i\epsilon) + c_4^{(1,1)}(y)
\end{aligned} \tag{A.3}$$

ndR11

with

$$\begin{aligned}
c_1^{(1,1)}(y, \lambda) = & (1-\chi) \frac{b'b''' - 2(b'')^2}{(b')^3} (b''\psi - \omega_{0k}), \\
c_2^{(1,1)}(y, \lambda) = & \left\{ \partial_y^2 \left(\frac{(1-\chi)b''}{b'} \right) - 2b'\partial_y \left[(1-\chi) \left(\frac{b'''}{(b')^2} - \frac{(b'')^2}{(b')^3} \right) \right] - (1-\chi)b'' \left[\frac{b'''}{(b')^2} - \frac{(b'')^2}{(b')^3} \right] \right\} \psi \\
& + \left[\frac{(1-\chi)b'''}{b'} - 2b'\partial_y \left(\frac{(1-\chi)b''}{(b')^2} \right) - \frac{(1-\chi)(b'')^2}{(b')^2} \right] D_\lambda \psi - (\partial_y^2 - k^2) \left(\frac{(1-\chi)\omega_{0k}}{b'} \right) \\
& + 2b'\partial_y \left[(1-\chi) \left(\frac{\omega'_{0k}}{(b')^2} - \frac{b''\omega_{0k}}{(b')^3} \right) \right] + (1-\chi)b'' \left[\frac{\omega'_{0k}}{(b')^2} - \frac{b''\omega_{0k}}{(b')^3} \right] + b'c_2^{(2,0)}, \\
c_1^{(1,1)}(y) = & \frac{2(1-\chi)b''}{b'}, \\
c_2^{(1,1)}(y) = & (1-\chi) \left[\frac{b'''}{b'} - \frac{2(b'')^2}{(b')^2} \right], \\
c_3^{(1,1)}(y, \lambda) = & b'c_3^{(2,0)} + (\partial_y^2 - k^2) \left[(1-\chi) \left(\frac{\omega'_{0k}}{(b')^2} - \frac{b''\omega_{0k}}{(b')^3} \right) \right] \\
& - \left\{ \partial_y^2 \left[(1-\chi) \left(\frac{b'''}{(b')^2} - \frac{(b'')^2}{(b')^3} \right) \right] - \frac{2k^2(1-\chi)(b'')^2}{(b')^3} \right\} \psi \\
& + \left\{ -2\partial_y \left[(1-\chi) \left(\frac{b'''}{(b')^2} - \frac{(b'')^2}{(b')^3} \right) \right] + \frac{(1-\chi)b''}{(b')^2} \left[\frac{b'''}{b'} - \frac{2(b'')^2}{(b')^2} \right] \right\} \partial_y \psi \\
& - \left\{ \partial_y^2 \left(\frac{(1-\chi)b''}{(b')^2} \right) + \frac{(1-\chi)b''}{(b')^2} \left[\frac{b'''}{b'} - \frac{2(b'')^2}{(b')^2} \right] \right\} D_\lambda \psi \\
& - \left[2\partial_y \left(\frac{(1-\chi)b''}{(b')^2} \right) + \frac{2(1-\chi)(b'')^2}{(b')^3} \right] \partial_y D_\lambda \psi, \\
c_4^{(1,1)}(y, \lambda) = & \chi \left[\frac{b''D_\lambda \psi + b'''\psi - \omega'_{0k}}{(b(y) - \lambda + i\epsilon)^2} + \frac{b'''\partial_\lambda \psi}{b(y) - \lambda + i\epsilon} \right] + 2b''\partial_y r_{2,0,\epsilon}^l + b'''\partial_y r_{2,0,\epsilon}^l.
\end{aligned}$$

$$(4) \text{ For } \mathcal{R}_{3,0,\epsilon}^l(y, \lambda) = \left(\partial_y^2 - k^2 - \frac{b''(y)}{b(y) - \lambda + i\epsilon} \right) (\partial_\lambda^3 \psi_{k,\epsilon}^l(y, \lambda) - \mathcal{S}_{3,\epsilon}^l):$$

$$\begin{aligned}
\mathcal{R}_{3,0,\epsilon}^l = & c_1^{(3,0)}(y) \frac{\log(b(y) - \lambda + i\epsilon)}{b(y) - \lambda + i\epsilon} + \frac{c_2^{(3,0)}(y) + c_1^{(3,0)}(y) \partial_y D_\lambda^2 \psi + c_2^{(3,0)}(y) \partial_y D_\lambda \psi + c_3^{(3,0)}(y) \partial_y \psi}{b(y) - \lambda + i\epsilon} \\
& + c_3^{(3,0)}(y) \log(b(y) - \lambda + i\epsilon) + c_4^{(3,0)}(y)
\end{aligned} \tag{A.4}$$

ndR30

with

$$c_1^{(3,0)}(y, \lambda) = (1-\chi) \left[\frac{3b''b^{(4)} + 6(b''')^2}{(b')^4} - \frac{42(b'')^2b'''}{(b')^5} + \frac{42(b'')^4}{(b')^6} \right] \psi$$

$$\begin{aligned}
& + (1 - \chi) \left[-\frac{3b^{(4)}}{(b')^4} + \frac{30b''b'''}{(b')^5} - \frac{42(b'')^3}{(b')^6} \right] \omega_{0k} + (1 - \chi) \left[-\frac{6b'''}{(b')^4} + \frac{12(b'')^2}{(b')^5} \right] \omega'_{0k} \\
& + (1 - \chi) \left[\frac{6b''b'''}{(b')^4} - \frac{12(b'')^3}{(b')^5} \right] D_\lambda \psi \\
c_2^{(3,0)}(y, \lambda) = & - \left\{ 2k^2 \partial_y \left[-\frac{(1 - \chi)b''}{(b')^3} \right] + \partial_y^2 \left[(1 - \chi) \left(\frac{2b'''}{(b')^3} - \frac{3(b'')^2}{(b')^4} \right) \right] \right. \\
& - k^2 \left(\frac{2b''}{b'} \right) \left(\frac{3(1 - \chi)b''}{(b')^3} \right) \\
& + 2b' \partial_y \left[(1 - \chi) \left(-\frac{3b^{(4)}}{(b')^4} + \frac{16b''b'''}{(b')^5} - \frac{15(b'')^3}{(b')^6} - \frac{k^2 b''}{(b')^4} \right) \right] \\
& \left. + b'' \left((1 - \chi) \left(-\frac{3b^{(4)}}{(b')^4} + \frac{16b''b'''}{(b')^5} - \frac{15(b'')^3}{(b')^6} - \frac{k^2 b''}{(b')^4} \right) \right) \right\} \psi \\
& - \left\{ \partial_y^2 \left[\frac{3(1 - \chi)b''}{(b')^3} \right] + \left(\frac{b'''}{b'} - \frac{2(b'')^2}{(b')^2} \right) \left(\frac{3(1 - \chi)b''}{(b')^3} \right) \right. \\
& + 2b' \partial_y \left[(1 - \chi) \left(-\frac{6b'''}{(b')^4} + \frac{12(b'')^2}{(b')^5} \right) \right] \\
& \left. + b'' \left((1 - \chi) \left(-\frac{6b'''}{(b')^4} + \frac{12(b'')^2}{(b')^5} \right) \right) \right\} D_\lambda \psi \\
& - \left\{ 2b' \partial_y \left[-\frac{3(1 - \chi)b''}{(b')^4} \right] + b'' \left(-\frac{3(1 - \chi)b''}{(b')^4} \right) \right\} D_\lambda^2 \psi \\
& - \left[\left(-\frac{2(1 - \chi)}{(b')^3} \right) + 2b' \left(\frac{3(1 - \chi)}{(b')^4} \right) \right] \omega'''_{0k} \\
& - \left\{ 2\partial_y \left[-\frac{2(1 - \chi)}{(b')^3} \right] + \left(\frac{3(1 - \chi)b''}{(b')^4} \right) \right. \\
& \left. + 2b' \left[\left(-\frac{12(1 - \chi)b''}{(b')^5} \right) + \partial_y \left[\frac{3(1 - \chi)}{(b')^4} \right] \right] + b'' \left(\frac{3(1 - \chi)}{(b')^4} \right) \right\} \omega''_{0k} \\
& - \left\{ \partial_y^2 \left[-\frac{2(1 - \chi)}{(b')^3} \right] - k^2 \left(-\frac{2(1 - \chi)}{(b')^3} \right) + 2\partial_y \left[\frac{3(1 - \chi)b''}{(b')^4} \right] \right. \\
& + 2b' \left[\partial_y \left[-\frac{12(1 - \chi)b''}{(b')^5} \right] + \left((1 - \chi) \left(\frac{k^2}{(b')^4} - \frac{4b'''}{(b')^5} + \frac{15(b'')^2}{(b')^6} \right) \right) \right] \\
& \left. + b'' \left(-\frac{12(1 - \chi)b''}{(b')^5} \right) \right\} \omega'_{0k} \\
& - \left\{ \partial_y^2 \left[\frac{3(1 - \chi)b''}{(b')^4} \right] - k^2 \left(\frac{3(1 - \chi)b''}{(b')^4} \right) \right. \\
& + 2b' \partial_y \left[(1 - \chi) \left(\frac{k^2}{(b')^4} - \frac{4b'''}{(b')^5} + \frac{15(b'')^2}{(b')^6} \right) \right] \\
& \left. + b'' \left((1 - \chi) \left(\frac{k^2}{(b')^4} - \frac{4b'''}{(b')^5} + \frac{15(b'')^2}{(b')^6} \right) \right) \right\} \omega_{0k}, \\
c_1^{(3,0)}(y) = & -2b' \left(-\frac{3(1 - \chi)b''}{(b')^4} \right), \\
c_2^{(3,0)}(y) = & -2\partial_y \left[\frac{3(1 - \chi)b''}{(b')^3} \right] - \left(\frac{2b''}{b'} \right) \left(\frac{3(1 - \chi)b''}{(b')^3} \right) - 2b' \left((1 - \chi) \left(-\frac{6b'''}{(b')^4} + \frac{12(b'')^2}{(b')^5} \right) \right),
\end{aligned}$$

$$\begin{aligned}
\mathfrak{c}_3^{(3,0)}(y) &= -\partial_y^2 \left[-\frac{(1-\chi)b''}{(b')^3} \right] - 2\partial_y \left[(1-\chi) \left(\frac{2b'''}{(b')^3} - \frac{3(b'')^2}{(b')^4} \right) \right] \\
&\quad + \left(\frac{b'''}{b'} - \frac{2(b'')^2}{(b')^2} \right) \left(\frac{3(1-\chi)b''}{(b')^3} \right) \\
&\quad - 2b' \left((1-\chi) \left(-\frac{3b^{(4)}}{(b')^4} + \frac{16b''b'''}{(b')^5} - \frac{15(b'')^3}{(b')^6} - \frac{k^2b''}{(b')^4} \right) \right), \\
c_3^{(3,0)}(y) &= - \left\{ \partial_y^2 \left[(1-\chi) \left(-\frac{3b^{(4)}}{(b')^4} + \frac{16b''b'''}{(b')^5} - \frac{15(b'')^3}{(b')^6} - \frac{k^2b''}{(b')^4} \right) \right] \right. \\
&\quad - k^2 \left(\frac{2b''}{b'} \right) \left((1-\chi) \left(-\frac{6b'''}{(b')^4} + \frac{12(b'')^2}{(b')^5} \right) \right) \\
&\quad \left. - k^2 \left(\frac{2b'''}{b'} - \frac{6(b'')^2}{(b')^2} \right) \left(-\frac{3(1-\chi)b''}{(b')^4} \right) \right\} \psi \\
&\quad - \left\{ 2\partial_y \left[(1-\chi) \left(-\frac{3b^{(4)}}{(b')^4} + \frac{16b''b'''}{(b')^5} - \frac{15(b'')^3}{(b')^6} - \frac{k^2b''}{(b')^4} \right) \right] \right. \\
&\quad - \left(\frac{b'''}{b'} - \frac{2(b'')^2}{(b')^2} \right) \left((1-\chi) \left(-\frac{6b'''}{(b')^4} + \frac{12(b'')^2}{(b')^5} \right) \right) \\
&\quad \left. - \left(\frac{b^{(4)}}{b'} - \frac{6b''b'''}{(b')^2} + \frac{6(b'')^3}{(b')^3} \right) \left(-\frac{3(1-\chi)b''}{(b')^4} \right) \right\} \partial_y \psi \\
&\quad - \left\{ \partial_y^2 \left[(1-\chi) \left(-\frac{6b'''}{(b')^4} + \frac{12(b'')^2}{(b')^5} \right) \right] \right. \\
&\quad + \left(\frac{b'''}{b'} - \frac{2(b'')^2}{(b')^2} \right) \left((1-\chi) \left(-\frac{6b'''}{(b')^4} + \frac{12(b'')^2}{(b')^5} \right) \right) \\
&\quad \left. + \left(\left(\frac{b^{(4)}}{b'} - \frac{6b''b'''}{(b')^2} + \frac{6(b'')^3}{(b')^3} \right) - 2k^2 \left(\frac{2b''}{b'} \right) \right) \left(-\frac{3(1-\chi)b''}{(b')^4} \right) \right\} D_\lambda \psi \\
&\quad - \left\{ 2\partial_y \left[(1-\chi) \left(-\frac{6b'''}{(b')^4} + \frac{12(b'')^2}{(b')^5} \right) \right] \right. \\
&\quad \left. + \left(\frac{2b''}{b'} \right) \left((1-\chi) \left(-\frac{6b'''}{(b')^4} + \frac{12(b'')^2}{(b')^5} \right) \right) \right\} \partial_y D_\lambda \psi \\
&\quad - \left\{ \partial_y^2 \left[-\frac{3(1-\chi)b''}{(b')^4} \right] + \left(\frac{2b'''}{b'} - \frac{6(b'')^2}{(b')^2} \right) \left(-\frac{3(1-\chi)b''}{(b')^4} \right) \right\} D_\lambda^2 \psi \\
&\quad - \left\{ 2\partial_y \left[-\frac{3(1-\chi)b''}{(b')^4} \right] + 2 \left(\frac{2b''}{b'} \right) \left(-\frac{3(1-\chi)b''}{(b')^4} \right) \right\} \partial_y D_\lambda^2 \psi \\
&\quad - \left(\frac{3(1-\chi)}{(b')^4} \right) \omega_{0k}^{(4)} \\
&\quad - \left(\left(-\frac{12(1-\chi)b''}{(b')^5} \right) + 2\partial_y \left[\frac{3(1-\chi)}{(b')^4} \right] \right) \omega_{0k}''' \\
&\quad - \left\{ 2\partial_y \left[-\frac{12(1-\chi)b''}{(b')^5} \right] \right. \\
&\quad \left. + \left((1-\chi) \left(\frac{k^2}{(b')^4} - \frac{4b'''}{(b')^5} + \frac{15(b'')^2}{(b')^6} \right) \right) \right\}
\end{aligned}$$

$$\begin{aligned}
& + \partial_y^2 \left[\frac{3(1-\chi)}{(b')^4} \right] - k^2 \left(\frac{3(1-\chi)}{(b')^4} \right) \Big\} \omega''_{0k} \\
& - \left\{ \partial_y^2 \left[-\frac{12(1-\chi)b''}{(b')^5} \right] - k^2 \left(-\frac{12(1-\chi)b''}{(b')^5} \right) \right. \\
& \quad \left. + 2\partial_y \left[(1-\chi) \left(\frac{k^2}{(b')^4} - \frac{4b'''}{(b')^5} + \frac{15(b'')^2}{(b')^6} \right) \right] \right\} \omega'_{0k} \\
& - \left\{ \partial_y^2 \left[(1-\chi) \left(\frac{k^2}{(b')^4} - \frac{4b'''}{(b')^5} + \frac{15(b'')^2}{(b')^6} \right) \right] \right. \\
& \quad \left. - k^2 \left((1-\chi) \left(\frac{k^2}{(b')^4} - \frac{4b'''}{(b')^5} + \frac{15(b'')^2}{(b')^6} \right) \right) \right\} \omega_{0k}, \\
c_4^{(3,0)}(y) = & \chi \left[\frac{6(b''\psi - \omega_{0k})}{(b(y) - \lambda + i\epsilon)^4} + \frac{6b''\partial_\lambda\psi}{(b(y) - \lambda + i\epsilon)^3} + \frac{3b''\partial_\lambda^2\psi}{(b(y) - \lambda + i\epsilon)^2} \right] \\
& + \chi' \left\{ -\frac{4(b''\psi - \omega_{0k})}{b'(b(y) - \lambda + i\epsilon)^3} + \frac{2}{(b(y) - \lambda + i\epsilon)^2} \left[\left(-\frac{b'''}{(b')^2} + \frac{(b'')^2}{(b')^3} \right) \psi \right. \right. \\
& \quad \left. \left. + \frac{2b''}{(b')^2} \partial_y\psi + \frac{\omega'_{0k}}{(b')^2} - \frac{b''\omega_{0k}}{(b')^3} - \frac{3b''}{(b')^2} D_\lambda\psi \right] \right\} \\
& + \chi'' \frac{b''\psi - \omega_{0k}}{(b')^2(b(y) - \lambda + i\epsilon)^2}.
\end{aligned}$$

$$(5) \text{ For } \mathcal{R}_{2,1,\epsilon}^t(y, \lambda) = \left(\partial_y^2 - k^2 - \frac{b''(y)}{b(y) - \lambda + i\epsilon} \right) (\partial_\lambda^2 D_\lambda \psi_{k,\epsilon}^t - \mathcal{S}_{2,1,\epsilon}^t):$$

$$\begin{aligned}
\mathcal{R}_{2,1,\epsilon}^t(y, \lambda) = & c_1^{(2,1)}(y) \frac{\log(b(y) - \lambda + i\epsilon)}{b(y) - \lambda + i\epsilon} + \frac{c_2^{(2,1)}(y) + c_1^{(2,1)}(y) \partial_y D_\lambda^2 \psi + c_2^{(2,1)}(y) \partial_y D_\lambda \psi + c_3^{(2,1)}(y) \partial_y \psi}{b(y) - \lambda + i\epsilon} \\
& + c_3^{(2,1)}(y) \log(b(y) - \lambda + i\epsilon) + c_4^{(2,1)}(y),
\end{aligned} \tag{A.5}$$

ndR21

with

$$\begin{aligned}
c_1^{(2,1)}(y, \lambda) = & b' c_1^{(3,0)} - \left[(1-\chi) \frac{2b'b'''' - 3(b'')^2}{(b')^4} \left(b'''' - \frac{2(b'')^2}{b'} \right) + \frac{(1-\chi)b''}{(b')^3} \left(b^{(4)} - \frac{6b''b'''}{b'} \right. \right. \\
& \left. \left. + \frac{6(b'')^3}{(b')^2} \right) \right] \psi - \frac{(1-\chi)b''}{(b')^3} \left(2b'''' - \frac{4(b'')^2}{b'} \right) D_\lambda \psi + \left(b'' \frac{3(1-\chi)b''}{(b')^4} \right. \\
& \left. + (1-\chi) \frac{2b'b'''' - 3(b'')^2}{(b')^4} - \frac{4b''}{b'} \frac{(1-\chi)b''}{(b')^3} \right) \omega'_{0k} \\
& + \left[(1-\chi) \frac{k^2(b')^2 b'' + (b')^2 b^{(4)} - 4b'b''b'''' + 3(b'')^3}{(b')^5} \right. \\
& \left. + b'' \left[-(1-\chi) \frac{k^2(b')^2 - b'b'''' + 3(b'')^2}{(b')^5} \right] - \frac{2b''}{b'} (1-\chi) \frac{2b'b'''' - 3(b'')^2}{(b')^4} \right. \\
& \left. - \left(\frac{2b''''}{b'} - \frac{6(b'')^2}{(b')^2} \right) \frac{(1-\chi)b''}{(b')^3} \right] \omega_{0k}, \\
c_2^{(2,1)}(y) = & b' c_2^{(3,0)} + \left(\frac{(1-\chi)b'''}{(b')^2} - 2b' \partial_y \left[\frac{(1-\chi)b''}{(b')^3} \right] - b'' \frac{(1-\chi)b''}{(b')^3} \right) D_\lambda^2 \psi \\
& + \left[\frac{(1-\chi)b''b'''}{(b')^3} + 2 \frac{(1-\chi)b''}{(b')^2} \left(\frac{b'''}{b'} - \frac{2(b'')^2}{(b')^2} \right) + 2\partial_y^2 \left[\frac{(1-\chi)b''}{(b')^2} \right] \right. \\
& \left. - 2b' \partial_y \left[(1-\chi) \frac{2b'b'''' - 3(b'')^2}{(b')^4} \right] - b'' (1-\chi) \frac{2b'b'''' - 3(b'')^2}{(b')^4} \right] D_\lambda \psi
\end{aligned}$$

$$\begin{aligned}
& + \left[\frac{k^2(1-\chi)b'''}{(b')^2} - 2k^2\partial_y \left[\frac{(1-\chi)b''}{(b')^2} \right] - \partial_y^2 \left[(1-\chi) \left(-\frac{b'''}{(b')^2} + \frac{(b'')^2}{(b')^3} \right) \right] \right. \\
& - \frac{4k^2b''}{b'} \frac{(1-\chi)b''}{(b')^2} - 2b'\partial_y \left[(1-\chi) \frac{k^2(b')^2b'' + (b')^2b^{(4)} - 4b'b''b''' + 3(b'')^3}{(b')^5} \right] \\
& - b''(1-\chi) \frac{k^2(b')^2b'' + (b')^2b^{(4)} - 4b'b''b''' + 3(b'')^3}{(b')^5} \Big] \psi + \left(-\frac{1-\chi}{(b')^2} \right. \\
& + 2b' \frac{1-\chi}{(b')^3} \omega_{0k}''' + (-2\partial_y \left[\frac{1-\chi}{(b')^2} \right] + \frac{(1-\chi)b''}{(b')^3} + 2b'\partial_y \left[\frac{1-\chi}{(b')^3} \right] \\
& - 2b' \frac{3(1-\chi)b''}{(b')^4} + b'' \frac{1-\chi}{(b')^3} \omega_{0k}'' + (k^2 \frac{1-\chi}{(b')^2} - \partial_y^2 \left[\frac{1-\chi}{(b')^2} \right] \\
& + 2\partial_y \left[\frac{(1-\chi)b''}{(b')^3} \right] - 2b'\partial_y \left[\frac{3(1-\chi)b''}{(b')^4} \right] \\
& - 2b' \left[-(1-\chi) \frac{k^2(b')^2 - b'b''' + 3(b'')^2}{(b')^5} \right] - b'' \frac{3(1-\chi)b''}{(b')^4} \omega_{0k}' + (\\
& - k^2 \frac{(1-\chi)b''}{(b')^3} + \partial_y^2 \left[\frac{(1-\chi)b''}{(b')^3} \right] \\
& - 2b'\partial_y \left[-(1-\chi) \frac{k^2(b')^2 - b'b''' + 3(b'')^2}{(b')^5} \right] \\
& \left. - b'' \left[-(1-\chi) \frac{k^2(b')^2 - b'b''' + 3(b'')^2}{(b')^5} \right] \right) \omega_{0k}, \\
\mathfrak{c}_1^{(2,1)}(y) &= b' \mathfrak{c}_1^{(3,0)} - 2b' \frac{(1-\chi)b''}{(b')^3}, \\
\mathfrak{c}_2^{(2,1)}(y) &= b' \mathfrak{c}_2^{(3,0)} - \frac{2(1-\chi)b'''}{(b')^2} + \frac{4b''}{b'} \frac{(1-\chi)b''}{(b')^2} + 4\partial_y \left[\frac{(1-\chi)b''}{(b')^2} \right] \\
& - 2b'(1-\chi) \frac{2b'b''' - 3(b'')^2}{(b')^4}, \\
\mathfrak{c}_3^{(2,1)}(y) &= b' \mathfrak{c}_3^{(3,0)} - \frac{(1-\chi)b''b'''}{(b')^3} - \partial_y^2 \left[\frac{(1-\chi)b''}{(b')^2} \right] - 2\partial_y \left[(1-\chi) \left(-\frac{b'''}{(b')^2} + \frac{(b'')^2}{(b')^3} \right) \right] \\
& - 2 \frac{(1-\chi)b''}{(b')^2} \left(\frac{b'''}{b'} - \frac{2(b'')^2}{(b')^2} \right) \\
& - 2b'(1-\chi) \frac{k^2(b')^2b'' + (b')^2b^{(4)} - 4b'b''b''' + 3(b'')^3}{(b')^5}, \\
c_3^{(2,1)}(y) &= b' c_3^{(3,0)} - \frac{3(1-\chi)b''}{(b')^4} (\omega_{0k}''' - k^2 \omega_{0k}') - 2\partial_y \left[\frac{3(1-\chi)b''}{(b')^4} \right] \omega_{0k}'' \\
& - \partial_y^2 \left[\frac{3(1-\chi)b''}{(b')^4} \right] \omega_{0k}' \\
& - 2\partial_y \left[(1-\chi) \frac{k^2(b')^2b'' + (b')^2b^{(4)} - 4b'b''b''' + 3(b'')^3}{(b')^5} \right] \partial_y \psi \\
& - \partial_y^2 \left[(1-\chi) \frac{k^2(b')^2b'' + (b')^2b^{(4)} - 4b'b''b''' + 3(b'')^3}{(b')^5} \right] \psi \\
& - \left[-(1-\chi) \frac{k^2(b')^2 - b'b''' + 3(b'')^2}{(b')^5} \right] (\omega_{0k}'' - k^2 \omega_{0k}) \\
& - 2\partial_y \left[-(1-\chi) \frac{k^2(b')^2 - b'b''' + 3(b'')^2}{(b')^5} \right] \omega_{0k}' \\
& - \partial_y^2 \left[-(1-\chi) \frac{k^2(b')^2 - b'b''' + 3(b'')^2}{(b')^5} \right] \omega_{0k}
\end{aligned}$$

$$\begin{aligned}
& - (1 - \chi) \frac{2b'b''' - 3(b'')^2}{(b')^4} \left[\left(\frac{b'''}{b'} - \frac{2(b'')^2}{(b')^2} \right) (D_\lambda \psi - \partial_y \psi) \right. \\
& + \frac{2b''}{b'} (\partial_y D_\lambda \psi - k^2 \psi) \left. \right] - 2\partial_y \left[(1 - \chi) \frac{2b'b''' - 3(b'')^2}{(b')^4} \right] \partial_y D_\lambda \psi \\
& - \partial_y^2 \left[(1 - \chi) \frac{2b'b''' - 3(b'')^2}{(b')^4} \right] D_\lambda \psi \\
& - \frac{4b''}{b'} \frac{(1 - \chi)b''}{(b')^3} \left[\frac{\partial_y D_\lambda^2 \psi - k^2 D_\lambda \psi + \left(-\frac{b'''}{b'} + \frac{2(b'')^2}{(b')^2} \right) (D_\lambda \psi - \partial_y \psi)}{b'} \right. \\
& \quad \left. - \frac{2b''}{b'} (\partial_y D_\lambda \psi - k^2 \psi) \right] \\
& - \frac{(1 - \chi)b''}{(b')^3} \left[\left(-\frac{4(b'')^2}{(b')^2} + \frac{2b'''}{b'} \right) (D_\lambda^2 \psi - \partial_y D_\lambda \psi) + \frac{2b'''}{b'} (\partial_y D_\lambda \psi - k^2 \psi) \right] \\
& + \frac{2(b'')^2}{(b')^2} \frac{(1 - \chi)b''}{(b')^3} \left[D_\lambda^2 \psi - 2\partial_y D_\lambda \psi + k^2 \psi + \frac{b''}{b'} (D_\lambda \psi - \partial_y \psi) \right] \\
& - \frac{(1 - \chi)b''}{(b')^3} \left(-\frac{2b''b'''}{(b')^2} + \frac{b^{(4)}}{b'} \right) (D_\lambda \psi - \partial_y \psi) - 2\partial_y \left[\frac{(1 - \chi)b''}{(b')^3} \right] \partial_y D_\lambda^2 \psi \\
& - \partial_y^2 \left[\frac{(1 - \chi)b''}{(b')^3} \right] D_\lambda^2 \psi + \frac{1 - \chi}{(b')^3} (\omega_{0k}^{(4)} - k^2 \omega_{0k}'') + 2\partial_y \left[\frac{1 - \chi}{(b')^3} \right] \omega_{0k}''' \\
& + \partial_y^2 \left[\frac{1 - \chi}{(b')^3} \right] \omega_{0k}'', \\
c_4^{(2,1)}(y) &= 2b'' \partial_y r_{3,0,\epsilon}^t + b''' r_{3,0,\epsilon}^t \\
& + \chi \left[\frac{2(b'' D_\lambda \psi + b''' \psi - \omega_{0k}')}{(b(y) - \lambda + i\epsilon)^3} + \frac{2(b'' \partial_\lambda D_\lambda \psi + b''' \partial_\lambda \psi)}{(b(y) - \lambda + i\epsilon)^2} + \frac{b''' \partial_\lambda^2 \psi}{b(y) - \lambda + i\epsilon} \right] \\
& + 2\chi' \left[-\frac{b'' D_\lambda \psi + b''' \psi - \omega_{0k}'}{b'} + \frac{b''(b'' \psi - \omega_{0k})}{(b')^2} \right] \frac{1}{(b(y) - \lambda + i\epsilon)^2}.
\end{aligned}$$

(6) For $\mathcal{R}_{1,2,\epsilon}^t(y, \lambda) = \left(\partial_y^2 - k^2 - \frac{b''(y)}{b(y) - \lambda + i\epsilon} \right) (\partial_\lambda D_\lambda^2 \psi_{k,\epsilon}^t - \mathcal{S}_{1,2,\epsilon}^t)$:

$$\begin{aligned}
\mathcal{R}_{1,2,\epsilon}^t(y, \lambda) &= c_1^{(1,2)}(y) \frac{\log(b(y) - \lambda + i\epsilon)}{b(y) - \lambda + i\epsilon} + \frac{c_2^{(1,2)}(y) + c_1^{(1,2)}(y) \partial_y D_\lambda^2 \psi + c_2^{(1,2)}(y) \partial_y D_\lambda \psi + c_3^{(1,2)}(y) \partial_y \psi}{b(y) - \lambda + i\epsilon} \\
&+ c_3^{(1,2)}(y) \log(b(y) - \lambda + i\epsilon) + c_4^{(1,2)}(y),
\end{aligned} \tag{A.6}$$

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with

$$\begin{aligned}
c_1^{(1,2)}(y, \lambda) &= -(b')^2 c_1^{(3,0)} + 2b' c_1^{(2,1)} + b'' c_1^{(2,0)}, \\
c_2^{(1,2)}(y) &= -(b')^2 c_2^{(3,0)} + 2b' c_2^{(2,1)} + b'' c_2^{(2,0)} \\
&+ 2k^2 b' \left[\partial_y \left(\frac{1 - \chi}{(b')^2} \right) (b'' \psi - \omega_{0k}) + \frac{1 - \chi}{(b')^2} (b''' \psi - \omega_{0k}') \right] + k^2 \frac{1 - \chi}{(b')^2} b'' (b'' \psi - \omega_{0k}), \\
c_1^{(1,2)}(y) &= -(b')^2 c_1^{(3,0)} + 2b' c_1^{(2,1)}, \\
c_2^{(1,2)}(y) &= -(b')^2 c_2^{(3,0)} + 2b' c_2^{(2,1)} + b'' c_1^{(2,0)}, \\
c_3^{(1,2)}(y) &= -(b')^2 c_3^{(3,0)} + 2b' c_3^{(2,1)} + b'' c_2^{(2,0)} + 2k^2 b' \frac{1 - \chi}{(b')^2} b'', \\
c_3^{(1,2)}(y) &= -(b')^2 c_3^{(3,0)} + 2b' c_3^{(2,1)} + b'' c_3^{(2,0)} \\
&+ k^2 \left[2\partial_y \left(\frac{(1 - \chi)b''}{(b')^2} \right) \partial_y \psi + \partial_y^2 \left(\frac{(1 - \chi)b''}{(b')^2} \right) \psi - \frac{1 - \chi}{(b')^2} \omega_{0k}'' \right] \\
&+ k^2 \left[-2\partial_y \left(\frac{1 - \chi}{(b')^2} \right) \omega_{0k}' + \left(k^2 \frac{1 - \chi}{(b')^2} - \partial_y^2 \left(\frac{1 - \chi}{(b')^2} \right) \right) \omega_{0k} \right],
\end{aligned}$$

$$\begin{aligned}
c_4^{(1,2)}(y) = & -2(b'')^2 r_{3,0,\epsilon}^t + 4b'' \partial_y r_{2,1,\epsilon}^t + 2b''' r_{2,1,\epsilon}^t + 2b''' \partial_y r_{2,0,\epsilon}^t + b^{(4)} r_{2,0,\epsilon}^t \\
& + \chi \left[\frac{b'' D_\lambda^2 \psi + 2b''' D_\lambda \psi + b^{(4)} \psi - \omega_{0k}''}{(b(y) - \lambda + i\epsilon)^2} + \frac{2b''' \partial_\lambda D_\lambda \psi + b^{(4)} \partial_\lambda \psi}{b(y) - \lambda + i\epsilon} \right] \\
& + 2\chi' \left[\frac{b'' \partial_y \partial_\lambda \psi + b''' \partial_\lambda \psi}{b(y) - \lambda + i\epsilon} + \frac{b''(b'' \psi - \omega_{0k})}{b'(b(y) - \lambda + i\epsilon)^2} \right] \\
& + \chi'' \frac{b'' \partial_\lambda \psi}{b(y) - \lambda + i\epsilon}.
\end{aligned}$$

In the degenerate case, we group by dimension counting. We use the shorthands

$$f = F_{k,\epsilon}^t(y, \lambda), \quad b = b(y), \quad \chi = \chi_{j,\delta}^*(y).$$

$$\begin{aligned}
R_{1,0}^* = & \chi \log\left(\frac{b - \lambda + i\epsilon}{c}\right) \left[\frac{4(b'')^2}{(b')^3} \partial_y f - \frac{6(b'')^3}{(b')^4} f \right] + \frac{\chi}{(b - \lambda + i\epsilon)} \left[-\frac{2b''}{b'} \partial_y f + \frac{3(b'')^2}{(b')^2} f \right] & \text{order } -2 \\
& + \chi' \log\left(\frac{b - \lambda + i\epsilon}{c}\right) \left[-\frac{2b''}{(b')^2} \partial_y f + \frac{4(b'')^2}{(b')^3} f \right] + \frac{\chi'}{(b - \lambda + i\epsilon)} \left[-\frac{2b''}{b'} f \right] \\
& + \chi'' \log\left(\frac{b - \lambda + i\epsilon}{c}\right) \left[-\frac{b''}{(b')^2} f \right] \\
& + \chi \log\left(\frac{b - \lambda + i\epsilon}{c}\right) \left[-\frac{2b'''}{(b')^2} \partial_y f + \frac{6b''b'''}{(b')^3} f \right] + \frac{\chi}{(b - \lambda + i\epsilon)} \left[-\frac{2b'''}{b'} f \right] & \text{order } -1 \\
& + \chi' \log\left(\frac{b - \lambda + i\epsilon}{c}\right) \left[-\frac{2b'''}{(b')^2} f \right] \\
& + \chi \log\left(\frac{b - \lambda + i\epsilon}{c}\right) \left[-\frac{b''''}{(b')^2} f \right] & \text{order } 0 \\
& + \partial_\lambda r + \chi \log\left(\frac{b - \lambda + i\epsilon}{c}\right) \left[\frac{b''}{(b')^2} r \right] & r\text{-terms} \\
& + (1 - \chi) \left[\frac{1}{(b - \lambda + i\epsilon)^2} [-b'' f] \right]. & 1 - \chi \text{ terms}
\end{aligned}$$

$$\begin{aligned}
R_{2,0}^* = & \frac{\chi \log\left(\frac{b - \lambda + i\epsilon}{c}\right)}{(b - \lambda + i\epsilon)} \left[\frac{4(b'')^3}{(b')^4} f \right] & \text{order } -4 \\
& + \chi \log\left(\frac{b - \lambda + i\epsilon}{c}\right) \left[\frac{8(b'')^2}{(b')^4} \partial_y (D_\lambda f) - \frac{20(b'')^3}{(b')^5} D_\lambda f - \frac{28(b'')^3}{(b')^5} \partial_y f + \frac{60(b'')^4}{(b')^6} f + \frac{4k^2(b'')^2}{(b')^4} f \right] \\
& + \frac{\chi}{(b - \lambda + i\epsilon)} \left[-\frac{4b''}{(b')^2} \partial_y (D_\lambda f) + \frac{10(b'')^2}{(b')^3} D_\lambda f + \frac{2(b'')^2}{(b')^3} \partial_y f - \frac{15(b'')^3}{(b')^4} f \right] \\
& + \chi' \log\left(\frac{b - \lambda + i\epsilon}{c}\right) \left[-\frac{4b''}{(b')^3} \partial_y (D_\lambda f) + \frac{12(b'')^2}{(b')^4} D_\lambda f + \frac{6(b'')^2}{(b')^4} \partial_y f - \frac{24(b'')^3}{(b')^5} f \right] \\
& + \frac{\chi'}{(b - \lambda + i\epsilon)} \left[-\frac{4b''}{(b')^2} D_\lambda f + \frac{2b''}{(b')^2} \partial_y f + \frac{2(b'')^2}{(b')^3} f \right] \\
& + \chi'' \log\left(\frac{b - \lambda + i\epsilon}{c}\right) \left[-\frac{2b''}{(b')^3} D_\lambda f + \frac{3(b'')^2}{(b')^4} f \right] + \frac{\chi''}{(b - \lambda + i\epsilon)} \left[\frac{b''}{(b')^2} f \right] \\
& + \frac{\chi \log\left(\frac{b - \lambda + i\epsilon}{c}\right)}{(b - \lambda + i\epsilon)} \left[-\frac{2b''b'''}{(b')^3} f \right] & \text{order } -3 \\
& + \chi \log\left(\frac{b - \lambda + i\epsilon}{c}\right) \left[-\frac{4b'''}{(b')^3} \partial_y (D_\lambda f) + \frac{16b''b'''}{(b')^4} D_\lambda f + \frac{26b''b'''}{(b')^4} \partial_y f - \frac{84b''''(b'')^2}{(b')^5} f \right]
\end{aligned}$$

$$\begin{aligned}
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{4b'''}{(b')^2} D_\lambda f - \frac{2b'''}{(b')^2} \partial_y f + \frac{16b''b'''}{(b')^3} f \right] \\
& + \chi' \log \left(\frac{b-\lambda+i\epsilon}{c} \right) \left[-\frac{4b'''}{(b')^3} D_\lambda f - \frac{4b'''}{(b')^3} \partial_y f + \frac{24b''b'''}{(b')^4} f \right] \\
& + \frac{\chi'}{(b-\lambda+i\epsilon)} \left[-\frac{2b'''}{(b')^2} f \right] + \chi'' \log \left(\frac{b-\lambda+i\epsilon}{c} \right) \left[-\frac{2b'''}{(b')^3} f \right] \\
& + \chi \log \left(\frac{b-\lambda+i\epsilon}{c} \right) \left[-\frac{2b''''}{(b')^3} D_\lambda f - \frac{4b''''}{(b')^3} \partial_y f + \frac{12(b'')^2}{(b')^4} f + \frac{18b''b''''}{(b')^4} f \right] \quad \text{order } -2 \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{3b''''}{(b')^2} f \right] + \chi' \log \left(\frac{b-\lambda+i\epsilon}{c} \right) \left[-\frac{4b''''}{(b')^3} f \right] \\
& + \chi \log \left(\frac{b-\lambda+i\epsilon}{c} \right) \left[-\frac{2b^{(5)}}{(b')^3} f \right] \quad \text{order } -1 \\
& + \partial_\lambda^2 r + \chi \log \left(\frac{b-\lambda+i\epsilon}{c} \right) \left[\frac{2b''}{(b')^3} D_\lambda r - \frac{7(b'')^2}{(b')^4} r + \frac{2b'''}{(b')^3} r \right] \quad r\text{-terms} \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{b''}{(b')^2} r \right] \\
& + (1-\chi) \left[\frac{1}{(b-\lambda+i\epsilon)^3} [-2b''f] + \frac{1}{(b-\lambda+i\epsilon)^2} [-2b''\partial_\lambda f] \right] \quad 1-\chi \text{ terms} \\
& + \chi' \left[\frac{1}{(b-\lambda+i\epsilon)^2} \left[-\frac{2b''}{b'} f \right] \right] \quad \chi' \text{ terms}
\end{aligned}$$

$$\begin{aligned}
R_{1,1}^* & = -2b''\partial_y \left(\partial_\lambda^2 f - \chi \frac{b''f}{(b')^2(b-\lambda+i\epsilon)} \right) \quad \text{cancellations} \\
& + \frac{\chi \log \left(\frac{b-\lambda+i\epsilon}{c} \right)}{(b-\lambda+i\epsilon)} \left[\frac{2(b'')^3}{(b')^3} f \right] \quad \text{order } -3 \\
& + \chi \log \left(\frac{b-\lambda+i\epsilon}{c} \right) \left[\frac{2(b'')^2}{(b')^3} \partial_y (D_\lambda f) - \frac{4(b'')^3}{(b')^4} D_\lambda f - \frac{14(b'')^3}{(b')^4} \partial_y f + \frac{24(b'')^4}{(b')^5} f + \frac{2k^2(b'')^2}{(b')^3} f \right] \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{2b''}{b'} \partial_y (D_\lambda f) + \frac{3(b'')^2}{(b')^2} D_\lambda f + \frac{2(b'')^2}{(b')^2} \partial_y f - \frac{6(b'')^3}{(b')^3} f \right] \\
& + \chi' \log \left(\frac{b-\lambda+i\epsilon}{c} \right) \left[-\frac{2b''}{(b')^2} \partial_y (D_\lambda f) + \frac{4(b'')^2}{(b')^3} D_\lambda f + \frac{4(b'')^2}{(b')^3} \partial_y f - \frac{12(b'')^3}{(b')^4} f \right] \\
& + \frac{\chi'}{(b-\lambda+i\epsilon)} \left[-\frac{2b''}{b'} D_\lambda f + \frac{2(b'')^2}{(b')^2} f \right] \\
& + \chi'' \log \left(\frac{b-\lambda+i\epsilon}{c} \right) \left[-\frac{b''}{(b')^2} D_\lambda f + \frac{2(b'')^2}{(b')^3} f \right] \\
& + \frac{\chi \log \left(\frac{b-\lambda+i\epsilon}{c} \right)}{(b-\lambda+i\epsilon)} \left[-\frac{b''b'''}{(b')^2} f \right] \quad \text{order } -2 \\
& + \chi \log \left(\frac{b-\lambda+i\epsilon}{c} \right) \left[-\frac{2b'''}{(b')^2} \partial_y (D_\lambda f) + \frac{5b''b'''}{(b')^3} D_\lambda f + \frac{13b''b'''}{(b')^3} \partial_y f - \frac{36b''''(b'')^2}{(b')^4} f \right] \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{3b''''}{b'} D_\lambda f - \frac{b''''}{b'} \partial_y f + \frac{9b''b''''}{(b')^2} f \right] + \chi \left[-b''''\partial_\lambda^2 f \right] \\
& + \frac{(1-\chi)}{(b-\lambda+i\epsilon)} \left[-b''''\partial_\lambda f \right] + (1-\chi) \left[-b''''\partial_\lambda^2 f \right]
\end{aligned}$$

$$\begin{aligned}
& + \chi' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{2b'''}{(b')^2} D_\lambda f - \frac{2b'''}{(b')^2} \partial_y f + \frac{12b''b'''}{(b')^3} f \right] \\
& + \frac{\chi'}{(b-\lambda+i\epsilon)} \left[-\frac{2b'''}{b'} f \right] \\
& + \chi'' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{b'''}{(b')^2} f \right] \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{b''''}{(b')^2} D_\lambda f - \frac{2b''''}{(b')^2} \partial_y f + \frac{6(b''')^2}{(b')^3} f + \frac{8b''b''''}{(b')^3} f \right] \quad \text{order } -1 \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{2b''''}{b'} f \right] + \chi' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{2b''''}{(b')^2} f \right] \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{b^{(5)}}{(b')^2} f \right] \quad \text{order } 0 \\
& + D_\lambda \partial_\lambda r + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[\frac{b''}{(b')^2} D_\lambda r - \frac{4(b'')^2}{(b')^3} r + \frac{b'''}{(b')^2} r \right] \quad r\text{-terms} \\
& + (1-\chi) \left[\frac{1}{(b-\lambda+i\epsilon)^2} \left[-b'' D_\lambda f - b''' f \right] \right] \quad 1-\chi \text{ terms}
\end{aligned}$$

$$\begin{aligned}
R_{3,0}^* &= \frac{\chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right)}{(b-\lambda+i\epsilon)} \left[\frac{12(b'')^3}{(b')^5} D_\lambda f \right] \quad \text{order } -6 \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[\frac{12(b'')^2}{(b')^5} \partial_y (D_\lambda^2 f) - \frac{48(b'')^3}{(b')^6} D_\lambda^2 f - \frac{132(b'')^3}{(b')^6} \partial_y (D_\lambda f) + \frac{360(b'')^4}{(b')^7} D_\lambda f \right. \\
& + \frac{12k^2(b'')^2}{(b')^5} D_\lambda f + \frac{6(b'')^3}{(b')^4} \partial_\lambda^2 f + \frac{48(b'')^3}{(b')^5} \partial_y \partial_\lambda f + \frac{180(b'')^4}{(b')^7} \partial_y f + \frac{8k^2(b'')^2}{(b')^5} \partial_y f \\
& \left. - \frac{630(b'')^5}{(b')^8} f - \frac{20k^2(b'')^3}{(b')^6} f \right] \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{6b''}{(b')^3} \partial_y (D_\lambda^2 f) + \frac{21(b'')^2}{(b')^4} D_\lambda^2 f + \frac{12(b'')^2}{(b')^4} \partial_y (D_\lambda f) - \frac{78(b'')^3}{(b')^5} D_\lambda f \right. \\
& \left. - \frac{12(b'')^3}{(b')^5} \partial_y f - \frac{2b''k^2}{(b')^3} \partial_y f + \frac{105(b'')^4}{(b')^6} f + \frac{7k^2(b'')^2}{(b')^4} f \right] \\
& + \chi' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{6b''}{(b')^4} \partial_y (D_\lambda^2 f) + \frac{24(b'')^2}{(b')^5} D_\lambda^2 f + \frac{24(b'')^2}{(b')^5} \partial_y (D_\lambda f) \right. \\
& \left. - \frac{120(b'')^3}{(b')^6} D_\lambda f - \frac{30(b'')^3}{(b')^6} \partial_y f - \frac{2b''k^2}{(b')^4} \partial_y f + \frac{180(b'')^4}{(b')^7} f + \frac{8k^2(b'')^2}{(b')^5} f \right] \\
& + \frac{\chi'}{(b-\lambda+i\epsilon)} \left[-\frac{6b''}{(b')^3} D_\lambda^2 f + \frac{6b''}{(b')^3} \partial_y (D_\lambda f) + \frac{6(b'')^2}{(b')^4} D_\lambda f - \frac{6(b'')^3}{(b')^5} f - \frac{4b''k^2}{(b')^3} f \right] \\
& + \chi'' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{3b''}{(b')^4} D_\lambda^2 f + \frac{12(b'')^2}{(b')^5} D_\lambda f - \frac{15(b'')^3}{(b')^6} f - \frac{b''k^2}{(b')^4} f \right] \\
& + \frac{\chi''}{(b-\lambda+i\epsilon)} \left[\frac{3b''}{(b')^3} D_\lambda f - \frac{b''}{(b')^3} \partial_y f - \frac{3(b'')^2}{(b')^4} f \right] \\
& + \frac{\chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right)}{(b-\lambda+i\epsilon)} \left[-\frac{6b''b'''}{(b')^4} D_\lambda f + \frac{24b''''(b'')^2}{(b')^5} f \right] \quad \text{order } -5 \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{6b''''}{(b')^4} \partial_y (D_\lambda^2 f) + \frac{30b''b'''}{(b')^5} D_\lambda^2 f + \frac{102b''b'''}{(b')^5} \partial_y (D_\lambda f) \right. \\
& \left. - \frac{420b''''(b'')^2}{(b')^6} D_\lambda f - \frac{18b''b'''}{(b')^4} \partial_y \partial_\lambda f + \frac{24b''''(b'')^2}{(b')^5} \partial_\lambda f - \frac{250b''''(b'')^2}{(b')^6} \partial_y f \right]
\end{aligned}$$

$$\begin{aligned}
& -\frac{2b'''k^2}{(b')^4}\partial_y f + \frac{1110b'''(b'')^3}{(b')^7}f + \frac{12b''b'''k^2}{(b')^5}f \Big] + \frac{\chi}{(b-\lambda+i\epsilon)} \Big[-\frac{6b'''}{(b')^3}D_\lambda^2 f \\
& -\frac{6b'''}{(b')^3}\partial_y(D_\lambda f) + \frac{66b''b'''}{(b')^4}D_\lambda f + \frac{14b''b'''}{(b')^4}\partial_y f - \frac{150b'''(b'')^2}{(b')^5}f - \frac{4b'''k^2}{(b')^3}f \Big] \\
& + \chi' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \Big[-\frac{6b'''}{(b')^4}D_\lambda^2 f - \frac{12b'''}{(b')^4}\partial_y(D_\lambda f) + \frac{96b''b'''}{(b')^5}D_\lambda f + \frac{32b''b'''}{(b')^5}\partial_y f \\
& -\frac{250b'''(b'')^2}{(b')^6}f - \frac{2b'''k^2}{(b')^4}f \Big] \\
& + \frac{\chi'}{(b-\lambda+i\epsilon)} \Big[-\frac{6b'''}{(b')^3}D_\lambda f + \frac{2b'''}{(b')^3}\partial_y f + \frac{8b''b'''}{(b')^4}f \Big] \\
& + \chi'' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \Big[-\frac{6b'''}{(b')^4}D_\lambda f + \frac{16b''b'''}{(b')^5}f \Big] + \frac{\chi''}{(b-\lambda+i\epsilon)} \Big[\frac{2b'''}{(b')^3}f \Big] \\
& + \frac{\chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right)}{(b-\lambda+i\epsilon)} \Big[-\frac{6(b''')^2}{(b')^4}f - \frac{3b''b''''}{(b')^4}f \Big] \quad \text{order } -4 \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \Big[-\frac{3b''''}{(b')^4}D_\lambda^2 f - \frac{12b''''}{(b')^4}\partial_y(D_\lambda f) + \frac{48(b''')^2}{(b')^5}D_\lambda f + \frac{72b''b''''}{(b')^5}D_\lambda f \\
& -\frac{6(b''')^2}{(b')^4}\partial_\lambda f - \frac{3b''b''''}{(b')^4}\partial_\lambda f + \frac{32(b'')^2}{(b')^5}\partial_y f + \frac{56b''b''''}{(b')^5}\partial_y f - \frac{330b''(b''')^2}{(b')^6}f \\
& -\frac{265b''''(b'')^2}{(b')^6}f - \frac{b''''k^2}{(b')^4}f \Big] \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \Big[-\frac{9b''''}{(b')^3}D_\lambda f - \frac{3b''''}{(b')^3}\partial_y f + \frac{20(b''')^2}{(b')^4}f + \frac{35b''b''''}{(b')^4}f \Big] \\
& + \chi' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \Big[-\frac{12b''''}{(b')^4}D_\lambda f - \frac{6b''''}{(b')^4}\partial_y f + \frac{32(b''')^2}{(b')^5}f + \frac{56b''b''''}{(b')^5}f \Big] \\
& + \frac{\chi'}{(b-\lambda+i\epsilon)} \Big[-\frac{2b''''}{(b')^3}f \Big] + \chi'' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \Big[-\frac{3b''''}{(b')^4}f \Big] \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \Big[-\frac{6b^{(5)}}{(b')^4}D_\lambda f - \frac{6b^{(5)}}{(b')^4}\partial_y f + \frac{40b''b^{(5)}}{(b')^5}f + \frac{60b''b''''}{(b')^5}f \Big] \quad \text{order } -3 \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \Big[-\frac{4b^{(5)}}{(b')^3}f \Big] + \chi' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \Big[-\frac{6b^{(5)}}{(b')^4}f \Big] \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \Big[-\frac{3b^{(6)}}{(b')^4}f \Big] \quad \text{order } -2 \\
& + \partial_\lambda^3 r \quad \text{\textit{r}-terms} \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \Big[\frac{3b''}{(b')^4}D_\lambda^2 r - \frac{24(b'')^2}{(b')^5}D_\lambda r + \frac{6b'''}{(b')^4}D_\lambda r + \frac{15(b'')^3}{(b')^6}r \\
& + \frac{3b''''}{(b')^4}r - \frac{16b''b'''}{(b')^5}r + \frac{b''k^2}{(b')^4}r \Big] \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \Big[-\frac{3b''}{(b')^3}D_\lambda r + \frac{b''}{(b')^3}\partial_y r + \frac{3(b'')^2}{(b')^4}r \Big] \\
& + \frac{\chi'}{(b-\lambda+i\epsilon)} \Big[\frac{2b''}{(b')^3}r \Big] \\
& + (1-\chi) \Big[\frac{1}{(b-\lambda+i\epsilon)^4}[-6b''f] + \frac{1}{(b-\lambda+i\epsilon)^3}[-6b''\partial_\lambda f] + \frac{1}{(b-\lambda+i\epsilon)^2}[-3b''\partial_\lambda^2 f] \Big] \quad 1-\chi \text{ terms}
\end{aligned}$$

$$\begin{aligned}
& + \chi' \left[\frac{1}{(b - \lambda + i\epsilon)^3} \left[-\frac{4b''}{b'} f \right] + \frac{1}{(b - \lambda + i\epsilon)^2} \left[-\frac{6b''}{(b')^2} D_\lambda f + \frac{4b''}{(b')^2} \partial_y f - \frac{2b'''}{(b')^2} f \right] \right] & \chi' \text{ terms} \\
& + \chi'' \left[\frac{1}{(b - \lambda + i\epsilon)^2} \left[\frac{b''}{(b')^2} f \right] \right]. & \chi'' \text{ terms}
\end{aligned}$$

$$R_{2,1}^* = -2b'' \partial_y (\partial_\lambda^3 f - \chi S_{3,0}) - b''' (\partial_\lambda^3 f - \chi S_{3,0}) - \frac{b'''}{(b - \lambda + i\epsilon)} (\partial_\lambda^2 f - \chi S_{2,0}) \quad \text{cancellations}$$

$$\begin{aligned}
& - \frac{8\chi(b'')^3}{(b')^3} \partial_\lambda^2 f \log \left(\frac{b - \lambda + i\epsilon}{c} \right) \\
& + \frac{\chi \log \left(\frac{b - \lambda + i\epsilon}{c} \right)}{(b - \lambda + i\epsilon)} \left[\frac{8(b'')^3}{(b')^4} D_\lambda f - \frac{22(b'')^4}{(b')^5} f \right] & \text{order } -5
\end{aligned}$$

$$\begin{aligned}
& + \chi \log \left(\frac{b - \lambda + i\epsilon}{c} \right) \left[\frac{10(b'')^2}{(b')^4} \partial_y (D_\lambda^2 f) - \frac{28(b'')^3}{(b')^5} D_\lambda^2 f - \frac{94(b'')^3}{(b')^5} \partial_y (D_\lambda f) + \frac{250(b'')^4}{(b')^6} D_\lambda f \right. \\
& \quad + \frac{8k^2(b'')^2}{(b')^4} D_\lambda f + \frac{28(b'')^4}{(b')^5} \partial_\lambda f + \frac{200(b'')^4}{(b')^6} \partial_y f + \frac{2k^2(b'')^2}{(b')^4} \partial_y f - \frac{540(b'')^5}{(b')^7} f \\
& \quad \left. - \frac{30k^2(b'')^3}{(b')^5} f \right]
\end{aligned}$$

$$\begin{aligned}
& + \frac{\chi}{(b - \lambda + i\epsilon)} \left[-\frac{4b''}{(b')^2} \partial_y (D_\lambda^2 f) + \frac{16(b'')^2}{(b')^3} D_\lambda^2 f + \frac{10(b'')^2}{(b')^3} \partial_y (D_\lambda f) \right. \\
& \quad \left. - \frac{63(b'')^3}{(b')^4} D_\lambda f - \frac{2(b'')^3}{(b')^3} \partial_\lambda f - \frac{12(b'')^3}{(b')^4} \partial_y f + \frac{90(b'')^4}{(b')^5} f + \frac{2k^2(b'')^2}{(b')^3} f \right] \\
& + \chi' \log \left(\frac{b - \lambda + i\epsilon}{c} \right) \left[-\frac{4b''}{(b')^3} \partial_y (D_\lambda^2 f) + \frac{18(b'')^2}{(b')^4} D_\lambda^2 f + \frac{18(b'')^2}{(b')^4} \partial_y (D_\lambda f) - \frac{96(b'')^3}{(b')^5} D_\lambda f \right. \\
& \quad \left. - \frac{24(b'')^3}{(b')^5} \partial_y f + \frac{150(b'')^4}{(b')^6} f + \frac{2k^2(b'')^2}{(b')^4} f \right]
\end{aligned}$$

$$+ \frac{\chi'}{(b - \lambda + i\epsilon)} \left[-\frac{4b''}{(b')^2} D_\lambda^2 f + \frac{2b''}{(b')^2} \partial_y (D_\lambda f) + \frac{8(b'')^2}{(b')^3} D_\lambda f - \frac{2(b'')^2}{(b')^3} \partial_y f - \frac{6(b'')^3}{(b')^4} f \right]$$

$$+ \chi'' \log \left(\frac{b - \lambda + i\epsilon}{c} \right) \left[-\frac{2b''}{(b')^3} D_\lambda^2 f + \frac{9(b'')^2}{(b')^4} D_\lambda f - \frac{12(b'')^3}{(b')^5} f \right]$$

$$+ \frac{\chi''}{(b - \lambda + i\epsilon)} \left[\frac{b''}{(b')^2} D_\lambda f - \frac{2(b'')^2}{(b')^3} f \right]$$

$$+ \frac{\chi \log \left(\frac{b - \lambda + i\epsilon}{c} \right)}{(b - \lambda + i\epsilon)} \left[-\frac{2b''b'''}{(b')^3} D_\lambda f + \frac{26b''b'''}{(b')^4} f \right] \quad \text{order } -4$$

$$\begin{aligned}
& + \chi \log \left(\frac{b - \lambda + i\epsilon}{c} \right) \left[-\frac{4b'''}{(b')^3} \partial_y (D_\lambda^2 f) + \frac{23b''b'''}{(b')^4} D_\lambda^2 f + \frac{64b''b'''}{(b')^4} \partial_y (D_\lambda f) \right. \\
& \quad \left. - \frac{285b''b'''}{(b')^5} D_\lambda f - \frac{22b''b'''}{(b')^4} \partial_\lambda f - \frac{251b''b'''}{(b')^5} \partial_y f + \frac{925b''b'''}{(b')^6} f + \frac{15b''b'''k^2}{(b')^4} f \right]
\end{aligned}$$

$$\begin{aligned}
& + \frac{\chi}{(b - \lambda + i\epsilon)} \left[-\frac{4b'''}{(b')^2} D_\lambda^2 f - \frac{6b'''}{(b')^2} \partial_y (D_\lambda f) + \frac{52b''b'''}{(b')^3} D_\lambda f \right. \\
& \quad \left. + \frac{2b''b'''}{(b')^2} \partial_\lambda f + \frac{12b''b'''}{(b')^3} \partial_y f - \frac{125b''b'''}{(b')^4} f \right]
\end{aligned}$$

$$+ \chi' \log \left(\frac{b - \lambda + i\epsilon}{c} \right) \left[-\frac{4b'''}{(b')^3} D_\lambda^2 f - \frac{8b'''}{(b')^3} \partial_y (D_\lambda f) + \frac{72b''b'''}{(b')^4} D_\lambda f + \frac{24b''b'''}{(b')^4} \partial_y f - \frac{200b''b'''}{(b')^5} f \right]$$

$$+ \frac{\chi'}{(b - \lambda + i\epsilon)} \left[-\frac{6b'''}{(b')^2} D_\lambda f + \frac{2b'''}{(b')^2} \partial_y f + \frac{8b''b'''}{(b')^3} f \right]$$

$$\begin{aligned}
& + \chi'' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{4b'''}{(b')^3} D_\lambda f + \frac{12b''b'''}{(b')^4} f \right] + \frac{\chi''}{(b-\lambda+i\epsilon)} \left[\frac{b'''}{(b')^2} f \right] \\
& + \frac{\chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right)}{(b-\lambda+i\epsilon)} \left[-\frac{2(b''')^2}{(b')^3} f - \frac{2b''b'''}{(b')^3} f \right] \quad \text{order } -3 \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{2b''''}{(b')^3} D_\lambda^2 f - \frac{8b''''}{(b')^3} \partial_y(D_\lambda f) + \frac{32(b''')^2}{(b')^4} D_\lambda f + \frac{48b''b''''}{(b')^4} D_\lambda f \right. \\
& \quad \left. + \frac{4b''b''''}{(b')^3} \partial_\lambda f + \frac{28(b''')^2}{(b')^4} \partial_y f + \frac{48b''b''''}{(b')^4} \partial_y f - \frac{264b''(b''')^2}{(b')^5} f - \frac{212b''''(b'')^2}{(b')^5} f \right] \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{7b''''}{(b')^2} D_\lambda f - \frac{2b''''}{(b')^2} \partial_y f + \frac{16(b''')^2}{(b')^3} f + \frac{28b''b''''}{(b')^3} f \right] \\
& + \chi' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{8b''''}{(b')^3} D_\lambda f - \frac{4b''''}{(b')^3} \partial_y f + \frac{24(b''')^2}{(b')^4} f + \frac{42b''b''''}{(b')^4} f \right] \\
& + \frac{\chi'}{(b-\lambda+i\epsilon)} \left[-\frac{2b''''}{(b')^2} f \right] + \chi'' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{2b''''}{(b')^3} f \right] \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{4b^{(5)}}{(b')^3} D_\lambda f - \frac{4b^{(5)}}{(b')^3} \partial_y f + \frac{30b''b^{(5)}}{(b')^4} f + \frac{45b''''b^{(5)}}{(b')^4} f \right] \quad \text{order } -2 \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{3b^{(5)}}{(b')^2} f \right] + \chi' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{4b^{(5)}}{(b')^3} f \right] \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{2b^{(6)}}{(b')^3} f \right] \quad \text{order } -1 \\
& + D_\lambda \partial_\lambda^2 r \quad \text{r-terms} \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[\frac{2b''}{(b')^3} D_\lambda^2 r - \frac{17(b'')^2}{(b')^4} D_\lambda r + \frac{4b''''}{(b')^3} D_\lambda r + \frac{34(b'')^3}{(b')^5} r + \frac{2b''''}{(b')^3} r - \frac{24b''b''''}{(b')^4} r \right] \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{b''}{(b')^2} D_\lambda r + \frac{2(b'')^2}{(b')^3} r - \frac{b''''}{(b')^2} r \right] \\
& + (1-\chi) \left[\frac{1}{(b-\lambda+i\epsilon)^3} \left[-2b'' D_\lambda f - 2b''' f \right] + \frac{1}{(b-\lambda+i\epsilon)^2} \left[-2b'' \partial_\lambda D_\lambda f - 2b''' \partial_\lambda f \right] \right] \quad 1-\chi \text{ terms} \\
& + \chi' \left[\frac{1}{(b-\lambda+i\epsilon)^2} \left[-\frac{2b''}{b'} D_\lambda f + \frac{2(b'')^2}{(b')^2} f - \frac{2b''''}{b'} f \right] \right] \quad \chi' \text{ terms}
\end{aligned}$$

$$\begin{aligned}
R_{1,2}^* & = -4b'' \partial_y (\partial_\lambda^2 D_\lambda f - \chi S_{2,1}) + 2(b'')^2 (\partial_\lambda^3 f - \chi S_{3,0}) - 2b'''' \partial_y (\partial_\lambda^2 f - \chi S_{2,0}) \quad \text{cancellations} \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[\frac{12(b'')^2}{(b')^3} \partial_y(D_\lambda^2 f) - \frac{36(b'')^3}{(b')^4} D_\lambda^2 f - \frac{60(b'')^3}{(b')^4} \partial_y(D_\lambda f) + \frac{216(b'')^4}{(b')^5} D_\lambda f \right. \\
& \quad - \frac{4(b'')^2}{(b')^2} \partial_y \partial_\lambda(D_\lambda f) + \frac{2(b'')^3}{(b')^2} \partial_\lambda^2 f + \frac{8(b'')^3}{(b')^3} \partial_y \partial_\lambda f + \frac{96(b'')^4}{(b')^5} \partial_y f - \frac{390(b'')^5}{(b')^6} f \\
& \quad \left. - \frac{2k^2(b'')^3}{(b')^4} f \right] \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{2b''}{b'} \partial_y(D_\lambda^2 f) + \frac{11(b'')^2}{(b')^2} D_\lambda^2 f + \frac{4(b'')^2}{(b')^2} \partial_y(D_\lambda f) \right. \\
& \quad \left. - \frac{42(b'')^3}{(b')^3} D_\lambda f - \frac{6(b'')^3}{(b')^3} \partial_y f + \frac{60(b'')^4}{(b')^4} f \right] \\
& + \chi' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{2b''}{(b')^2} \partial_y(D_\lambda^2 f) + \frac{12(b'')^2}{(b')^3} D_\lambda^2 f + \frac{8(b'')^2}{(b')^3} \partial_y(D_\lambda f) \right]
\end{aligned}$$

$$\begin{aligned}
& -\frac{60(b'')^3}{(b')^4}D_\lambda f - \frac{12(b'')^3}{(b')^4}\partial_y f + \frac{96(b'')^4}{(b')^5}f] \\
& + \frac{\chi'}{(b-\lambda+i\epsilon)}\left[-\frac{2b''}{b'}D_\lambda^2 f + \frac{4(b'')^2}{(b')^2}D_\lambda f - \frac{4(b'')^3}{(b')^3}f\right] \\
& + \chi''\log\left(\frac{b-\lambda+i\epsilon}{c}\right)\left[-\frac{b''}{(b')^2}D_\lambda^2 f + \frac{4(b'')^2}{(b')^3}D_\lambda f - \frac{6(b'')^3}{(b')^4}f\right] \\
& + \frac{\chi\log\left(\frac{b-\lambda+i\epsilon}{c}\right)}{(b-\lambda+i\epsilon)}\left[-\frac{2b''b'''}{(b')^2}D_\lambda f + \frac{4b''''(b'')^2}{(b')^3}f\right] \quad \text{order } -3 \\
& + \chi\log\left(\frac{b-\lambda+i\epsilon}{c}\right)\left[-\frac{2b'''}{(b')^2}\partial_y(D_\lambda^2 f) + \frac{14b''b'''}{(b')^3}D_\lambda^2 f + \frac{44b''b'''}{(b')^3}\partial_y(D_\lambda f) \right. \\
& \quad - \frac{216b''''(b'')^2}{(b')^4}D_\lambda f - \frac{2b''b'''}{(b')^2}\partial_\lambda D_\lambda f - \frac{6b''b'''}{(b')^2}\partial_y\partial_\lambda f + \frac{4b''''(b'')^2}{(b')^3}\partial_\lambda f \\
& \quad \left. - \frac{126b''''(b'')^2}{(b')^4}\partial_y f + \frac{632b''''(b'')^3}{(b')^5}f\right] \\
& + \frac{\chi}{(b-\lambda+i\epsilon)}\left[-\frac{2b'''}{b'}D_\lambda^2 f - \frac{4b''''}{b'}\partial_y(D_\lambda f) + \frac{38b''b'''}{(b')^2}D_\lambda f - 2b'''\partial_\lambda D_\lambda f + \frac{6b''b'''}{(b')^2}\partial_y f \right. \\
& \quad \left. - \frac{88b''''(b'')^2}{(b')^3}f\right] + \chi\left[-2b'''\partial_\lambda^2 D_\lambda f\right] \\
& + \frac{(1-\chi)}{(b-\lambda+i\epsilon)}\left[-2b'''\partial_\lambda D_\lambda f\right] + (1-\chi)\left[-2b'''\partial_\lambda^2 D_\lambda f\right] + \chi'\log\left(\frac{b-\lambda+i\epsilon}{c}\right)\left[-\frac{2b'''}{(b')^2}D_\lambda^2 f \right. \\
& \quad \left. - \frac{4b''''}{(b')^2}\partial_y(D_\lambda f) + \frac{44b''b'''}{(b')^3}D_\lambda f + \frac{12b''b'''}{(b')^3}\partial_y f - \frac{126b''''(b'')^2}{(b')^4}f\right] \\
& + \frac{\chi'}{(b-\lambda+i\epsilon)}\left[-\frac{4b''''}{b'}D_\lambda f + \frac{6b''b'''}{(b')^2}f\right] + \chi''\log\left(\frac{b-\lambda+i\epsilon}{c}\right)\left[-\frac{2b'''}{(b')^2}D_\lambda f + \frac{6b''b'''}{(b')^3}f\right] \\
& + \frac{\chi\log\left(\frac{b-\lambda+i\epsilon}{c}\right)}{(b-\lambda+i\epsilon)}\left[-\frac{2(b''')^2}{(b')^2}f - \frac{b''b''''}{(b')^2}f\right] \quad \text{order } -2 \\
& + \chi\log\left(\frac{b-\lambda+i\epsilon}{c}\right)\left[-\frac{b''''}{(b')^2}D_\lambda^2 f - \frac{4b''''}{(b')^2}\partial_y(D_\lambda f) + \frac{16(b''')^2}{(b')^3}D_\lambda f + \frac{32b''b''''}{(b')^3}D_\lambda f \right. \\
& \quad - \frac{2(b''')^2}{(b')^2}\partial_\lambda f - \frac{b''b''''}{(b')^2}\partial_\lambda f + \frac{16(b''')^2}{(b')^3}\partial_y f + \frac{24b''b''''}{(b')^3}\partial_y f - \frac{162b''(b''')^2}{(b')^4}f \\
& \quad \left. - \frac{138b''''(b'')^2}{(b')^4}f\right] + \frac{\chi}{(b-\lambda+i\epsilon)}\left[-\frac{4b''''}{b'}D_\lambda f - b''''\partial_\lambda f - \frac{2b''''}{b'}\partial_y f + \frac{14(b''')^2}{(b')^2}f \right. \\
& \quad \left. + \frac{19b''b''''}{(b')^2}f\right] + \chi\left[-b''''\partial_\lambda^2 f\right] + \frac{(1-\chi)}{(b-\lambda+i\epsilon)}\left[-b''''\partial_\lambda f\right] + (1-\chi)\left[-b''''\partial_\lambda^2 f\right] \\
& + \chi'\log\left(\frac{b-\lambda+i\epsilon}{c}\right)\left[-\frac{4b''''}{(b')^2}D_\lambda f - \frac{2b''''}{(b')^2}\partial_y f + \frac{16(b''')^2}{(b')^3}f + \frac{24b''b''''}{(b')^3}f\right] \\
& + \frac{\chi'}{(b-\lambda+i\epsilon)}\left[-\frac{2b''''}{b'}f\right] + \chi''\log\left(\frac{b-\lambda+i\epsilon}{c}\right)\left[-\frac{b''''}{(b')^2}f\right] \\
& + \chi\log\left(\frac{b-\lambda+i\epsilon}{c}\right)\left[-\frac{2b^{(5)}}{(b')^2}D_\lambda f - \frac{2b^{(5)}}{(b')^2}\partial_y f + \frac{18b''b^{(5)}}{(b')^3}f + \frac{24b''b''''}{(b')^3}f\right] \quad \text{order } -1 \\
& + \frac{\chi}{(b-\lambda+i\epsilon)}\left[-\frac{2b^{(5)}}{b'}f\right] + \chi'\log\left(\frac{b-\lambda+i\epsilon}{c}\right)\left[-\frac{2b^{(5)}}{(b')^2}f\right] \\
& + \chi\log\left(\frac{b-\lambda+i\epsilon}{c}\right)\left[-\frac{b^{(6)}}{(b')^2}f\right] \quad \text{order } 0
\end{aligned}$$

$$\begin{aligned}
& + D_\lambda^2 \partial_\lambda r + \chi \log \left(\frac{b - \lambda + i\epsilon}{c} \right) \left[\frac{b''}{(b')^2} D_\lambda^2 r - \frac{4(b'')^2}{(b')^3} D_\lambda r + \frac{2b'''}{(b')^2} D_\lambda r + \frac{6(b'')^3}{(b')^4} r \right. \\
& \quad \left. + \frac{b''''}{(b')^2} r - \frac{6b''b'''}{(b')^3} r \right] \quad r\text{-terms} \\
& + (1 - \chi) \left[\frac{1}{(b - \lambda + i\epsilon)^2} \left[-b'' D_\lambda^2 f - 2b''' D_\lambda f - b'''' f \right] \right]. \quad 1 - \chi \text{ terms}
\end{aligned}$$

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